Question 9

Q: In an application like facial recognition, explain how different levels of Computer Vision are involved from image acquisition to decision making.

Theoretical Concept: Levels of Computer Vision

Low-Level: Capturing the face, adjusting brightness, and filtering noise.

Mid-Level: Extracting features (edges, corners) and isolating the face from the background (segmentation/Haar Cascades).

High-Level: Using machine learning to match the extracted facial topology against a database to make a final authentication decision.

OpenCV Python Solution

```
import cv2
import matplotlib.pyplot as plt
from google.colab import files

# Upload Image
print("Upload a face image")
uploaded = files.upload()

filename = list(uploaded.keys())[0]

# Read Image
img = cv2.imread(filename)
rgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)

# Load Haar Cascade Face Detector
face_detector = cv2.CascadeClassifier(
    cv2.data.haarcascades + "haarcascade_frontalface_default.xml"
)

gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
```

```python
# Detect Faces
faces = face_detector.detectMultiScale(
    gray,
    scaleFactor=1.2,
    minNeighbors=5
)

# Draw Face Box
for (x, y, w, h) in faces:
    cv2.rectangle(rgb, (x, y), (x+w, y+h), (0,255,0), 2)
    cv2.putText(rgb,
                "Face Detected",
                (x, y-10),
                cv2.FONT_HERSHEY_SIMPLEX,
                0.8,
                (255,0,0),
                2)

# Display Image
plt.figure(figsize=(7,7))
plt.imshow(rgb)
plt.title("Facial Recognition")
plt.axis("off")
plt.show()

# Decision Making
print("\n========== COMPUTER VISION REPORT ==========")
print("Low-Level Vision  : Image Captured")
print("Mid-Level Vision  : Face Detected")

if len(faces) > 0:
    print("High-Level Vision : Face Present")
    print("Decision          : Authorized User (Demo)")
else:
    print("High-Level Vision : No Face Found")
    print("Decision          : Access Denied")

print("============================================")
```

Expected Output

The output shows a highly speckled, noisy image next to a perfectly restored, flat gray image. The median filter calculates out extreme outliers without blurring actual structure.